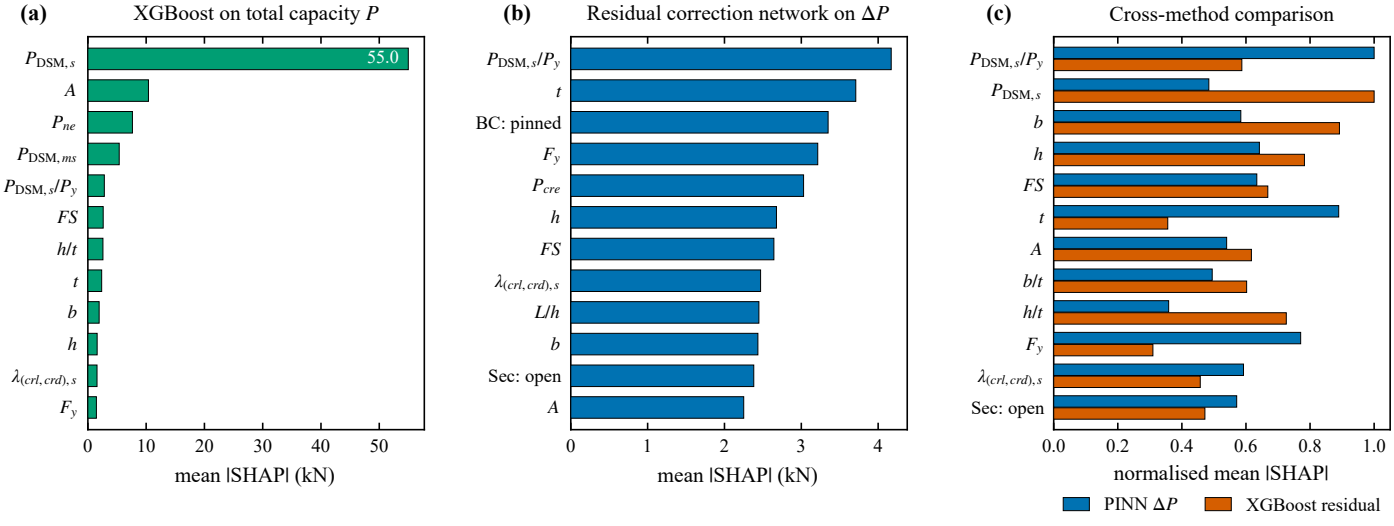
(a)
XGBoost on total capacity P
$P_{\mathrm{DSM},s}$
A
P_{ne}
$P_{\mathrm{DSM},ms}$
$P_{\mathrm{DSM},s}/P_y$
FS
h/t
t
b
h
$\lambda_{(crl,crd),s}$
F_y
55.0
0
10
20
30
40
50
mean |SHAP| (kN)
(b)
Residual correction network on ΔP
$P_{\mathrm{DSM},s}/P_y$
t
BC: pinned
F_y
P_{cre}
h
FS
$\lambda_{(crl,crd),s}$
L/h
b
Sec: open
A
0
1
2
3
4
mean |SHAP| (kN)
(c)
Cross-method comparison
$P_{\mathrm{DSM},s}/P_y$
$P_{\mathrm{DSM},s}$
b
h
FS
t
A
b/t
h/t
F_y
$\lambda_{(crl,crd),s}$
Sec: open
0.0
0.2
0.4
0.6
0.8
1.0
normalised mean |SHAP|
PINN ΔP
XGBoost residual